0580_w20_qp_42_Q3

3 **(a)** Beth invests \$2000 at a rate of 2% per year compound interest.

(i) Calculate the value of this investment at the end of 5 years.

\$ ..... [2]

(ii) Calculate the overall percentage increase in the value of Beth's investment at the end of 5 years.

..... % [2]

(iii) Calculate the minimum number of complete years it takes for the value of Beth's investment to increase from \$2000 to more than \$2500.

..... [3]

(b) The population of a village decreases exponentially at a rate of 4% each year.
The population is now 255.

Calculate the population 16 years ago.

..... [3]

0580_w20_ms_42_Q3

Question	Answer	Marks	Partial Marks
3(a)(i)	2210 or 2208 or 2208.2, or 2208.16…	**2**	**M1** for $2000 \times \left(1+\frac{2}{100}\right)^5$ oe
3(a)(ii)	10.4 or 10.5 or 10.40 to 10.41	**2**	**M1** for $\frac{\textit{their}\,\textbf{(a)(i)} - 2000}{2000}$ [×100] or $\frac{\textit{their}\,\textbf{(a)(i)}}{2000} \times 100$ or $\left(1+\frac{2}{100}\right)^5 - 1$ or $\left(1+\frac{2}{100}\right)^5 \times 100$ oe
3(a)(iii)	12	**3**	**B2** for 11.3 or 11.26 to 11.27 **OR** **M2** for [2000 ×] $\left(1+\frac{2}{100}\right)^{11}$ oe or [2000 ×] $\left(1+\frac{2}{100}\right)^{12}$ oe seen or **M1** for [2000 ×] $\left(1+\frac{2}{100}\right)^{n}$ oe, $n > 5$ oe or for $2000 \times \left(1+\frac{2}{100}\right)^{n}$ = or > or ⩾ 2500 oe

3(b)	490 cao	**3**	**M2** for $p \times \left(1-\frac{4}{100}\right)^{16} = 255$ oe soi by 490.0... or **M1** for $p \times \left(1-\frac{4}{100}\right)^{n} = 255$ oe, $n > 1$ oe